

Task 1:

Common BIOS/POST Error Messages and Causes

Objective: Identify five common BIOS/POST error messages displayed during system startup and describe the underlying hardware or configuration issues associated with each.

- **CMOS Checksum Error - Defaults Loaded**

Indicated Issue: Depleted/dead CMOS battery (CR2032 coin cell) on the motherboard, or corrupted BIOS configuration settings requiring a battery replacement and BIOS reconfiguration.

- **NO BOOT DEVICE AVAILABLE / Operating System Not Found**

Indicated Issue: The system cannot locate a bootable drive (HDD/SSD), caused by a loose/disconnected SATA/NVMe cable, drive hardware failure, or incorrect boot priority order in the BIOS.

- **CPU Fan Error**

Indicated Issue: The CPU cooling fan is either not spinning, unplugged from the CPU_FAN motherboard header, or failing, posing an immediate CPU overheating risk.

- **Keyboard Error or No Keyboard Present**

Indicated Issue: Physical disconnection of the keyboard, a damaged USB/PS2 port, or a failed input device check during POST initialization.

- **Memory Size Decreased**

Indicated Issue: Failure of one or more RAM modules to initialize, caused by improperly seated memory sticks, dirty RAM slots, or a defective RAM module.

Task 2:

AMI BIOS 3-Beep POST Code Diagnosis

Objective: Analyze a 3-beep POST code scenario on an AMI BIOS system and outline effective hardware troubleshooting steps.

1. Beep Code Meaning:

- **BIOS Type:** AMI (American Megatrends Inc.) BIOS
 - **Pattern Observed:** 3 Short Beeps (Repeating, No Display)
 - **Diagnosis: Base 64K RAM Memory Failure (Memory Read/Write Error).** This indicates that the system BIOS cannot read or initialize the first 64KB block of System Memory (RAM).
-

2. Recommended Troubleshooting Steps:

1. Reseat and Clean Memory Modules:

- Power down the PC, unplug the power cable, and open the cabinet.
- Remove all RAM sticks from their slots and clean the golden contact pins using a soft eraser or isopropyl alcohol.
- Clear any dust from the DIMM slots using compressed air and firmly reinsert the RAM until the side clips lock automatically.

2. Isolate Faulty RAM Stick or DIMM Slot:

- Test the computer with **only one RAM stick** inserted at a time to identify if a specific RAM module is defective.
- If a single RAM module still triggers 3 beeps, try inserting it into a different DIMM slot to rule out a damaged motherboard slot.

Task 3:

BIOS Setup Utility Inspection and System Identification

Task 4:

CMOS Checksum Error Analysis and Resolution

Objective: Explain the root cause of the "CMOS checksum error – Defaults loaded" POST error message and provide a step-by-step resolution process.

1. Meaning of the Error:

The "**CMOS checksum error – Defaults loaded**" message indicates that the BIOS/UEFI firmware's non-volatile memory (CMOS) has lost stored configuration data or calculated an invalid checksum hash during POST initialization.

This happens when the **CR2032 3V Lithium coin battery** on the motherboard is fully discharged or failing. Because the CMOS memory loses constant low-voltage power, all user-defined settings (date/time, drive boot order, XMP

profiles) are wiped, and the BIOS automatically restores its factory default values to allow system booting.

2. Resolution Steps:

1. Replace the CMOS Battery:

- Power off the PC and disconnect the main power cable.
- Open the side panel, locate the CR2032 coin-cell battery on the motherboard, press the retention clip to release it, and install a brand-new **CR2032 (3V) battery**.

2. Reconfigure and Save BIOS Settings:

- Plug in the power cord and turn on the system.
- Press DEL or F2 during POST to enter the BIOS Setup.

- Reset the current **System Date and Time**, configure the correct **Boot Order/Priority**, and re-enable any specific settings (e.g., AHCI mode, XMP/DOCP).
- Press **F10** to Save and Exit BIOS. Verify that the PC boots normally without showing the error again.